EE502 - Linear Systems Theory **Spring 2013**

Lecture 9

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9.1 External Input-Output Stability

9.1.1 Signal Norms

In the context of dynamical systems, we classify signals primarily based on their domain:

- **Continuous Time (CT)** bilateral signals are mappings defined by $f : \mathbb{R} \mapsto \mathbb{R}^n$ (or for the unilateral case $f : \mathbb{R}^{\geq 0} \mapsto \mathbb{R}^n$).
- **Discrete Time (DT)** bilateral signals are mappings defined by $g : \mathbb{Z} \mapsto \mathbb{R}^n$ (or for the unilateral case $g : \mathbb{Z}^{\geq 0} \mapsto \mathbb{R}^n$).

Graphical Examples:

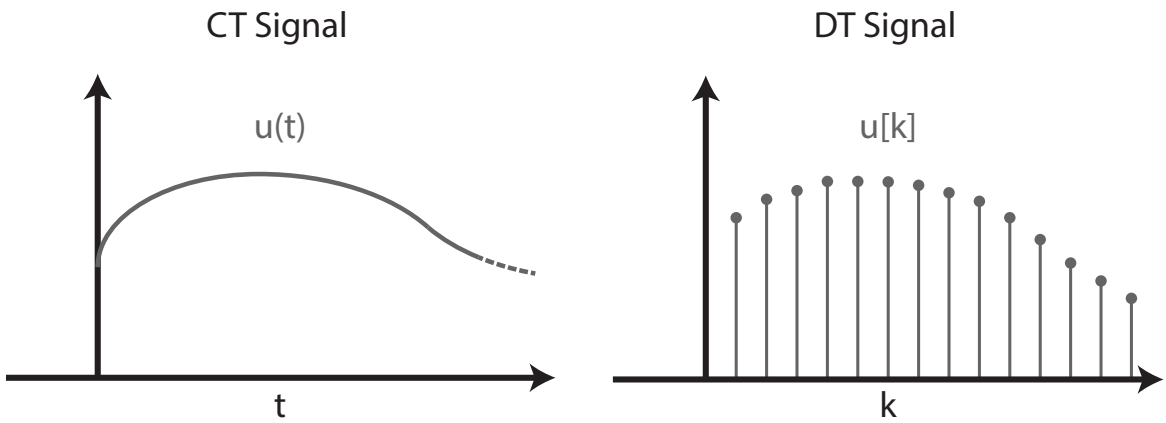

Figure 9.1: CT vs DT Signal Concept

∞-norm (Peak Magnitude)

In the characterization and analysis of input-output stability of linear dynamical systems, the most commonly used norm concept is the ∞-norm, which is technically a measure of peak magnitude over time. In practical engineering terms, this is often associated with **strict safety constraints** (e.g., maximum voltage, peak temperature, or physical limits).

For scalar signals, the ∞-norm is defined as:

$$\begin{aligned} \|f\|_\infty &\triangleq \sup_k |f(k)| \quad \text{(DT)} \\ &\triangleq \sup_t |f(t)| \quad \text{(CT)} \end{aligned}$$

The "sup" denotes the *supremum* or *least upper bound*, the value that is approached arbitrarily closely but never (i.e., at any finite time) exceeded. Note that this naturally extends the standard ∞-norm

definition for finite-dimensional vectors applied to the infinite-dimensional case. Let's recall the ∞-norm of an n-dimensional vector:

$$\|v\|_\infty \triangleq \max_{i\in[1,n]} |v_i| \ , \text{ where } v \in \mathbb{R}^n$$

A scalar signal, $f(\cdot)$, is called **bounded** if $\|f\|_\infty = M < \infty$. This is the fundamental signal measure adopted in Bounded-Input Bounded-Output (BIBO) stability.

For multi-variate signals, we add a new "dimension" in addition to the time dimension. Thus, we evaluate the peak vector norm over all time:

$$\begin{aligned} \|f\|_\infty &\triangleq \sup_k \|f(k)\|_\infty \quad \text{(DT)} \\ &\triangleq \sup_t \|f(t)\|_\infty \quad \text{(CT)} \end{aligned}$$

The spaces of all signals with a finite ∞-norm are denoted by ℓ_∞ and $\mathcal{L}_\infty$ for DT and CT signals, respectively.

Examples: ∞-norms of some CT and DT unilateral signals (i.e., $t \geq 0$, $k \geq 0$):

$f(t) = 1, \quad \|f\|_\infty = 1$	$g[k] = 1, \quad \|g\|_\infty = 1$
$f(t) = t, \quad \|f\|_\infty = \infty$	$g[k] = k, \quad \|g\|_\infty = \infty$
$f(t) = e^t, \quad \|f\|_\infty = \infty$	$g[k] = 2^k, \quad \|g\|_\infty = \infty$
$f(t) = 1 - e^{-t}, \quad \|f\|_\infty = 1$	$g[k] = 1 - 0.5^k, \quad \|g\|_\infty = 1$
$f(t) = \delta(t), \quad \|f\|_\infty = \infty$	$g[k] = \delta[k], \quad \|g\|_\infty = 1$

2-norm (Energy)

The 2-norm is the most fundamental signal measure in optimal control theory ($\mathcal{H}_2$ control), and it represents the square root of the **"total energy"** of the signal. For scalar signals, the 2-norm is defined as:

$$\begin{aligned} \|f\|_2 &\triangleq \left[\sum_k (f[k])^2\right]^{\frac{1}{2}} \quad \text{(DT)} \\ &\triangleq \left[\int_{-\infty}^{\infty} (f(t))^2 dt\right]^{\frac{1}{2}} \quad \text{(CT)} \end{aligned}$$

Signals belonging to the ℓ_2 or $\mathcal{L}_2$ spaces are often called *finite energy* signals. Note that many practical signals (like constant signals or persistent sine waves) have finite power but infinite energy, meaning they do not belong to this space.

For multivariate signals, we adopt the standard vector 2-norm inner product:

$$\|f\|_2 \triangleq \left[\sum_k \|f[k]\|_2^2\right]^{\frac{1}{2}} \quad \text{(DT)} \qquad \|f\|_2 \triangleq \left[\int_{-\infty}^{\infty} \|f(t)\|_2^2 \, dt\right]^{\frac{1}{2}} \quad \text{(CT)}$$

Examples:

$f(t) = 1, \quad \|f\|_2 = \infty$	$g[k] = 1, \quad \|g\|_2 = \infty$
$f(t) = e^{-t}, \quad \|f\|_2 = 1/\sqrt{2}$	$g[k] = 0.5^k, \quad \|g\|_2 = 2/\sqrt{3}$
$f(t) = \delta(t), \quad \|f\|_2 = 1$	$g[k] = \delta[k], \quad \|g\|_2 = 1$

1-norm (Action/Total Accumulation)

The 1-norm of a signal is referred to as the "action" of the signal. Practically, it measures absolute total accumulation over time. In control engineering, this often models cumulative quantities like **total fuel consumed** by thrusters or total absolute error.

$$\|f\|_1 \triangleq \left[\sum_k |f[k]|\right] \quad \text{(DT)} \qquad\qquad \|f\|_1 \triangleq \left[\int_{-\infty}^{\infty} |f(t)|\, dt\right] \quad \text{(CT)}$$

In order to generalize the 1-norm for multi-variate signals, we adopt the 1-norm definition of vectors ($\|v\|_1 \triangleq \sum_{i=1}^{n} |v_i|$):

$$\|f\|_1 \triangleq \sum_k \|f[k]\|_1 \quad \text{(DT)} \qquad\qquad \|f\|_1 \triangleq \int_{-\infty}^{\infty} \|f(t)\|_1\, dt \quad \text{(CT)}$$

Examples:

$$f(t) = e^{-t}, \quad \|f\|_1 = 1 \qquad\qquad g[k] = 0.5^k, \quad \|g\|_1 = 2$$
$$f(t) = \delta(t), \quad \|f\|_1 = 1 \qquad\qquad g[k] = \delta[k], \quad \|g\|_1 = 1$$

p-norm (Generalization)

The p-norm generalizes the previous cases. Let $p \geq 1$; the p-norm for vector-valued signals is defined as:

$$\|f\|_p \triangleq \left[\sum_k \|f[k]\|_p^p\right]^{1/p} \quad \text{(DT)} \qquad\qquad \|f\|_p \triangleq \left[\int_{-\infty}^{\infty} \|f(t)\|_p^p\, dt\right]^{1/p} \quad \text{(CT)}$$

9.1.2 Input-Output Stability

The most important notion of input-output stability in the analysis of dynamical systems is termed ℓ_p (or $\mathcal{L}_p$) stability, or simply p-stability.

Definition 9.1 (p-stability)**.** A system mapping an input signal u to an output signal y is ℓ_p stable (or p-stable) if there exists a finite constant $M \in \mathbb{R}$ such that:

$$\|y\|_p \leq M \|u\|_p\ , \quad \forall u \in \ell_p \,(\text{or } \mathcal{L}_p)$$

In other words, a p-stable system guarantees that an input with a finite p-norm will yield an output that also has a finite p-norm. In this context, the smallest possible bounding constant M is defined as the **induced system norm**, which essentially represents the system's maximum "gain" or amplification factor:

$$\|S\|_{p\text{-ind}} = \sup_{u \neq 0} \frac{\|y\|_p}{\|u\|_p}$$

BIBO Stability & $\mathcal{L}_\infty$-stability for LTI Systems

If we directly analyze the definition of p-stability, we can see that BIBO (Bounded-Input Bounded-Output) stability is exactly equivalent to p-stability where $p = \infty$. Let's formalize this for SISO LTI systems.

Theorem 9.2. *Let S_h be a CT-LTI SISO system characterized by its impulse response $h(t)$:*

$$y(t) = (h * u)(t) = \int_{-\infty}^{\infty} h(\tau)u(t-\tau)\,d\tau$$

S_h is BIBO stable (∞-stable) if and only if its impulse response is absolutely integrable. That is, $\exists M < \infty$ such that $\|h\|_1 = \int_{-\infty}^{\infty} |h(t)|\,dt = M$.

Proof. Sufficiency ($\Rightarrow$): Assume the impulse response is absolutely integrable ($\|h\|_1 = M < \infty$) and the input is bounded ($\|u\|_\infty = C < \infty$). We need to show the output $y(t)$ is also bounded.

$$|y(t)| = \left| \int_{-\infty}^{\infty} h(\tau)u(t-\tau)\,d\tau \right|$$

By applying the triangle inequality for integrals, we move the absolute value inside:

$$\leq \int_{-\infty}^{\infty} |h(\tau)u(t-\tau)|\,d\tau$$

Because $u(t)$ is bounded, we know that $|u(t-\tau)| \leq \|u\|_\infty$ for all t and τ. Substituting this worst-case bound yields:

$$\begin{aligned} &\leq \int_{-\infty}^{\infty} |h(\tau)|\,\|u\|_\infty\,d\tau \\ &= \|u\|_\infty \left[\int_{-\infty}^{\infty} |h(\tau)|\,d\tau \right], \quad \forall t \end{aligned}$$

Recognizing the integral as $\|h\|_1$, we obtain our final bound:

$$\|y\|_\infty \leq \|h\|_1\,\|u\|_\infty = MC < \infty$$

Since this bound holds uniformly for all t, the output is guaranteed to be bounded.

Necessity ($\Leftarrow$): We must prove that if $\|h\|_1 = \infty$, there exists at least one bounded input that produces an unbounded output.

Let $\|h\|_1 = \infty$. We construct a malicious input u designed specifically to drive the system to infinity at a given time t. We choose:

$$u(t-\tau) = \text{sign}(h(\tau))$$

Remark 9.3 (Insight on the Malicious Input)**.** Because the sign$(\cdot)$ function only outputs values in $\{-1, 0, 1\}$, this specific input is strictly bounded, satisfying $\|u\|_\infty \leq 1$. However, it perfectly aligns its phase with the impulse response, forcing every term in the convolution integral to become positive and act additively. It is the absolute "worst-case" scenario for the system.

Applying this bounded input yields:

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} h(\tau)\text{sign}(h(\tau))\,d\tau \\ &= \int_{-\infty}^{\infty} |h(\tau)|\,d\tau \\ &= \infty \end{aligned}$$

The output diverges to infinity, proving the system is unstable. □

Consequently, the least upper bound of the infinity norm of the output (the induced ∞-norm) is exactly the 1-norm of the impulse response:

$$\|S_h\|_{\infty\text{-ind}} = \sup_{\|u\|_\infty \leq 1} \|y\|_\infty = \|h\|_1$$

Theorem 9.4. *Let $S_{\mathcal{H}}$ be a CT-LTI MIMO system with m inputs, p outputs, and an impulse response matrix $\mathcal{H}(t)$. $S_{\mathcal{H}}$ is BIBO stable if and only if:*

$$\max_{i\in[1,p]} \sum_{j=1}^{m} \|h_{ij}\|_1 = M < \infty$$

Proof. For a MIMO system, the entire system is only as stable as its internal pathways. Let $h_{ij}(t)$ denote the impulse response from the jth input to the ith output. We analyze an arbitrary output terminal $y_i(t)$:

$$|y_i(t)| = \left| \sum_{j=1}^{m} \int_{-\infty}^{\infty} h_{ij}(\tau) u_j(t-\tau)\, d\tau \right|$$

Applying the triangle inequality to both the sum and the integral:

$$\leq \sum_{j=1}^{m} \int_{-\infty}^{\infty} |h_{ij}(\tau)|\, |u_j(t-\tau)|\, d\tau$$

Bounding each input channel by the global input norm $\|u\|_\infty$:

$$\leq \sum_{j=1}^{m} \int_{-\infty}^{\infty} |h_{ij}(\tau)|\, \|u\|_\infty\, d\tau$$

$$= \left[\sum_{j=1}^{m} \|h_{ij}\|_1 \right] \|u\|_\infty$$

This shows the ith output is bounded by the sum of the 1-norms of its respective input pathways. Maximizing over all possible output branches $i \in [1, p]$ yields the induced norm for the MIMO system:

$$\|S_{\mathcal{H}}\|_{\infty\text{-ind}} = \max_{i\in[1,p]} \left[\sum_{j=1}^{m} \|h_{ij}\|_1 \right]$$

□

Theorem 9.5 (Pole-Zero Interpretation)**.** *A finite-dimensional CT-LTI (or DT-LTI) system is BIBO (∞-stable) $\iff$*

- *Poles of the transfer function $\mathcal{H}(s)$ (or $\mathcal{G}(z)$) are located strictly in the Open Left-Half Plane (or strictly inside the unit circle for DT systems).*

Remark 9.6 (Theoretical Link)**.** Why does a finite 1-norm ($\|h\|_1 < \infty$) strictly require poles in the left-half plane? For a finite-dimensional LTI system, $h(t)$ is composed of linear combinations of natural modes such as $t^k e^{p_i t}$, where p_i are the system poles. The integral $\int_0^\infty \left| t^k e^{p_i t} \right| dt$ converges if and only if the real part of p_i is strictly negative, forcing the impulse response to exponentially decay to zero over time.

Example: Infinite-Dimensional System (Time-Delay)

Let us solidify the universality of the $\mathcal{L}_1$-norm condition with a system where simply checking a finite set of poles is insufficient. Consider an infinite-dimensional CT-LTI system representing an acoustic environment with an infinite series of echoes (a delay-feedback system). The dynamics are governed by:

$$y(t) = u(t) + \alpha y(t - T_d)$$

where $T_d > 0$ is the time delay and $\alpha \in \mathbb{R}$ is the reflection/attenuation factor.

To find the corresponding impulse response $h(t)$, we can elegantly derive it in the s-domain. Taking the Laplace transform of the system equation (assuming zero initial conditions) gives:

$$\begin{aligned} Y(s) &= U(s) + \alpha e^{-sT_d} Y(s) \\ H(s) &= \frac{Y(s)}{U(s)} = \frac{1}{1 - \alpha e^{-sT_d}} \end{aligned}$$

By expanding this transfer function using the infinite geometric series formula $\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$ (valid for the region of convergence), we get:

$$H(s) = \sum_{k=0}^{\infty} \left(\alpha e^{-sT_d}\right)^k = \sum_{k=0}^{\infty} \alpha^k e^{-skT_d}$$

Taking the inverse Laplace transform, and recalling the time-shift property $\mathcal{L}^{-1}\{e^{-as}\} = \delta(t - a)$, we arrive at the identical impulse response:

$$h(t) = \sum_{k=0}^{\infty} \alpha^k \delta(t - kT_d)$$

Now, we want to analyze the BIBO (∞-) stability of this system by evaluating its impulse response $\mathcal{L}_1$-norm:

$$\begin{aligned} \|h\|_1 &= \int_{-\infty}^{\infty} \left| \sum_{k=0}^{\infty} \alpha^k \delta(t - kT_d) \right| dt \\ &= \sum_{k=0}^{\infty} |\alpha|^k \int_{-\infty}^{\infty} \delta(t - kT_d)\, dt \\ &= \sum_{k=0}^{\infty} |\alpha|^k \end{aligned}$$

Case 1: $|\alpha| < 1$ (Attenuating Echoes)
If the reflection coefficient is strictly less than 1 (meaning the signal loses energy with each bounce), the geometric series converges:

$$\|h\|_1 = \frac{1}{1 - |\alpha|} < \infty$$

Since $\|h\|_1$ is finite, the system is **BIBO stable**. The induced ∞-norm (maximum peak amplification) is exactly $\frac{1}{1-|\alpha|}$.

Case 2: $|\alpha| \geq 1$ (Sustained or Growing Echoes)
If the echoes do not decay, the geometric series diverges ($\|h\|_1 = \infty$), and the system is **not BIBO stable**.

Remark 9.7 (Why finite-dimensional theorems fail here)**.** Taking the Laplace transform of the impulse response yields the transfer function $H(s) = \frac{1}{1-\alpha e^{-sT_d}}$. This is not a rational polynomial; it has an *infinite* number of poles located at $s = \frac{\ln(|\alpha|)+j(\angle\alpha+2\pi m)}{T_d}$ for $m \in \mathbb{Z}$. Standard stability checks designed for finite-dimensional systems (like Routh-Hurwitz) do not directly apply to this transcendental equation, perfectly showcasing why the $\|h\|_1 < \infty$ condition is the most fundamental and universally rigorous definition of BIBO stability.

$\mathcal{L}_1$-stability and Young's Inequality

Let's focus again on CT SISO systems. Suppose we want to find a bound C such that $\|y\|_1 \leq C\,\|u\|_1$ for all inputs $u \in \mathcal{L}_1$. From our previous derivation, we know that the absolute value of the output at any given time is bounded by the convolution of the absolute values:

$$|y(t)| \leq \int_{-\infty}^{\infty} |h(t-\tau)|\,|u(\tau)|\,d\tau$$

To find the $\mathcal{L}_1$ norm of the output, we integrate both sides over all time t:

$$\|y\|_1 = \int_{-\infty}^{\infty} |y(t)|\,dt \leq \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} |h(t-\tau)|\,|u(\tau)|\,d\tau \right) dt$$

By applying Tonelli's/Fubini's Theorem, we can safely swap the order of integration:

$$\leq \int_{-\infty}^{\infty} |u(\tau)| \left(\int_{-\infty}^{\infty} |h(t-\tau)|\,dt \right) d\tau$$

We can use a change of variables in the inner integral ($\gamma = t - \tau$, $d\gamma = dt$). Due to time-invariance, the shift by τ does not change the integral over infinite bounds:

$$\begin{aligned} &= \int_{-\infty}^{\infty} |u(\tau)| \left(\int_{-\infty}^{\infty} |h(\gamma)|\,d\gamma \right) d\tau \\ &= \|h\|_1 \int_{-\infty}^{\infty} |u(\tau)|\,d\tau \\ \|y\|_1 &\leq \|h\|_1\,\|u\|_1 \end{aligned}$$

This elegant result implies that if $\|h\|_1 = M < \infty$, the system is inherently 1-stable, and its induced 1-norm is bounded by M.

Remark 9.8 (Tightness of the 1-norm bound)**.** The bound derived above, $\|y\|_1 \leq \|h\|_1\,\|u\|_1$, is actually a **tight bound** (the least upper bound). We can prove this by evaluating an input that achieves equality. If we apply a unit Dirac delta impulse $u(t) = \delta(t)$, which has a 1-norm of $\|u\|_1 = 1$, the output is exactly the impulse response $y(t) = h(t)$. Therefore, $\|y\|_1 = \|h\|_1$, perfectly satisfying the equality condition. Thus, the induced 1-norm of the system is exactly $\|S\|_{\text{1-ind}} = \|h\|_1$.

Theorem 9.9 (Young's Convolution Inequality)**.** *The property derived above is actually a specific instance of a broader and highly fundamental mathematical theorem known as* ***Young's Convolution Inequality****. It states that for any $p \geq 1$, if $\|h\|_1 < \infty$ and $\|u\|_p < \infty$, then the output p-norm is bounded by:*

$$\|y\|_p = \|h * u\|_p \leq \|h\|_1\,\|u\|_p$$

Thus, a system with an absolutely integrable impulse response ($\|h\|_1 < \infty$) is not just BIBO (∞) stable, but is guaranteed to be ℓ_p-stable for *any* choice of $p \geq 1$.

Remark 9.10 (Conservativeness of Young's Inequality)**.** While Young's Convolution Inequality is a powerful mathematical tool to guarantee that stability exists, it generally provides a **conservative (loose) upper bound** for intermediate values of p (i.e., $1 < p < \infty$).

$\mathcal{L}_2$-stability and the $\mathcal{H}_\infty$-Norm

Another crucially important induced norm in linear systems and optimal control theory is the induced 2-norm, which measures the maximum **energy gain** of a system. By definition, for a stable LTI system:

$$\|S\|_{2\text{-ind}} = \sup_{\|u\|_2 \neq 0} \frac{\|y\|_2}{\|u\|_2}$$

While the induced ∞-norm ($\|S\|_{\infty\text{-ind}}$) is computed in the time domain via the 1-norm of the impulse response ($\|h\|_1$), the induced 2-norm is best understood and computed in the **frequency domain**.

To bridge the time and frequency domains, we rely on a fundamental result in harmonic analysis that establishes the conservation of energy:

Theorem 9.11 (Parseval's Theorem)**.** *For any finite-energy CT signal $f \in \mathcal{L}_2$, the total energy computed in the time domain is equal to the total energy computed in the frequency domain (scaled by $1/2\pi$):*

$$\|f\|_2^2 = \int_{-\infty}^{\infty} |f(t)|^2 \, dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(j\omega)|^2 \, d\omega$$

where $F(j\omega)$ is the continuous Fourier transform of $f(t)$.

Using Parseval's Theorem, we can elegantly analyze the energy transfer of a CT-LTI SISO system. We know that convolution in the time domain, $y(t) = (h * u)(t)$, becomes multiplication in the frequency domain:

$$Y(j\omega) = H(j\omega)U(j\omega)$$

We want to find the maximum possible energy of the output, $\|y\|_2^2$, for a given input energy, $\|u\|_2^2$. Let us evaluate $\|y\|_2^2$ step-by-step:

$$\begin{aligned}
\|y\|_2^2 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |Y(j\omega)|^2 \, d\omega && \text{(by Parseval's Theorem)} \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} |H(j\omega)U(j\omega)|^2 \, d\omega && \text{(System dynamics)} \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} |H(j\omega)|^2 \, |U(j\omega)|^2 \, d\omega && \text{(Property of absolute values)}
\end{aligned}$$

To bound this integral, we can pull the largest possible value of $|H(j\omega)|^2$ outside the integral. We define this maximum amplification factor as the supremum over all frequencies:

$$\leq \left(\sup_{\omega \in \mathbb{R}} |H(j\omega)|^2 \right) \frac{1}{2\pi} \int_{-\infty}^{\infty} |U(j\omega)|^2 \, d\omega$$

Recognizing the remaining integral as the input energy $\|u\|_2^2$ via Parseval's Theorem, and pulling the square outside the supremum:

$$= \left(\sup_{\omega \in \mathbb{R}} |H(j\omega)| \right)^2 \|u\|_2^2$$

Taking the square root of both sides gives our final bound:

$$\|y\|_2 \leq \left(\sup_{\omega \in \mathbb{R}} |H(j\omega)| \right) \|u\|_2$$

This inequality is strictly tight. It is achievable by concentrating the input energy $U(j\omega)$ in an infinitesimally narrow frequency band exactly centered around the frequency ω^* where the magnitude $|H(j\omega)|$ reaches its maximum peak. Therefore, the induced 2-norm is exactly the supremum of the frequency response magnitude.

Theorem 9.12 ($\mathcal{H}_\infty$-Norm)**.** *For a stable CT-LTI SISO system with frequency response $H(j\omega)$, the induced 2-norm is called the $\mathcal{H}_\infty$-**norm** (Hardy space infinity norm), defined as:*

$$\|S\|_{2\text{-}ind} = \|H\|_\infty \triangleq \sup_{\omega \in \mathbb{R}} |H(j\omega)|$$

For a MIMO system with frequency response matrix $\mathcal{H}(j\omega)$, the $\mathcal{H}_\infty$-norm is the supremum of the maximum singular value, $\bar{\sigma}$, over all frequencies:

$$\|\mathcal{H}\|_\infty \triangleq \sup_{\omega \in \mathbb{R}} \bar{\sigma}(\mathcal{H}(j\omega))$$

Remark 9.13 (Physical Interpretation)**.** Physically, the $\mathcal{H}_\infty$-norm represents the worst-case steady-state gain of the system to harmonic inputs. On a standard Bode magnitude plot, $\|H\|_\infty$ is simply the highest peak of the magnitude curve (the resonant peak). Minimizing this peak is the central objective of $\mathcal{H}_\infty$ **robust control**, as it limits the maximum amplification of finite-energy disturbances.